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MLA02

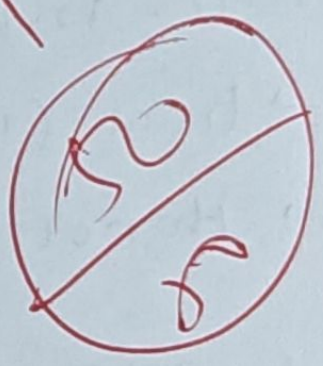
CIA-1.

Fundamental of M.L

Given data: $P(C) = 0.03$, sensitivity $P(+/C) = 0.90$,

Specificity $P(-/-C) = 0.94$;

D.p $\Rightarrow \lambda(no/C) = 50$, H.P $\Rightarrow \lambda(-C) = 2$



Correct decisions East = 0.

priors & likelihoods:

$P(-C) = 0.97$;

$P(+/C) = 0.90$;

$P(+/C) = 1 - 0.94 = 0.06$.

$P(-/c) = 0.10$

$P(-/-c) = 0.94$.

Evidence: $p(+) = 0.90 \times 0.03 + 0.06 \times 0.97$

$= 0.2070 + 0.0582 = 0.0852$

$P(-) = 1 - 0.0852 = 0.9148$.

Posteriors: $P(C/+) = 0.0270/0.852 = 0.5169$

$P(-C/+) = 0.6831$

$P(C/-) = (0.10 \times 0.03)/0.9148$

$= 0.033$;

$P(-C/-) = 0.9967$

Conditional risks:

$R(\text{treat}/+) = 2 \times 0.6831 = 1.3662$

$R(\text{treat no}/+) = 50 \times 0.3169 = 15.845$

$R(\text{treat}/-) = 2 \times 0.9967 = 1.9934$

$R(\text{no treat}/-) = 0.1640$

- minimum-risk decision:

  + → treat ($1.3662 < 15.845$);

  − → do not treat ($0.1640 < 1.9934$)

- Bayes risk: $R = 0.0852 \times 1.3662 + 0.9148 \times 0.1640$

  $= 0.1164 + 0.1500$

  $= 0.2664$ units per patient

- 2nd independent + test:

  posterior odds $= (0.3169/0.6831) \times (0.90/0.6) =$

  $= 0.4639 \times 15 = 6.959$

  $P(C/++) = 6.959/7.959 = 0.8744.$

- minimum-error (0-1 loss) rule:

  decide C only if $P(C/x) > 0.5$.

  since $P(C/+) = 0.3169 < 0.5$,

  real loss matrix $= 50 \times 0.03 = 1.5$ units p.p.

Interpretation & conclusion: A positive results give a 31.7% chance of disease, yet treatment is justified because a missed case costs 25 times a false alarm

2. Given data:

P(x,y) of load X (Low, medium, High) and Fault y (No, yes), Model $\phi = (0.5, 0.3, 0.2)$

All logarithms are base 2.

Load X	Y = NO	Y = yes	P(X)
Low	0.30	0.02	0.32
medium	0.28	0.10	0.38
high	0.10	0.20	0.30
P(Y)	0.68	0.32	1.00

Numerical Soln's:

- marginals: $p(x) = (0.32, 0.38, 0.30)$;

$p(y) = (0.68, 0.32)$

$H(X) = -\sum \log_2 P = 0.5260 + 0.5304 + 0.5211$

$= 1.5776$ bits

$H(Y) = 0.3784 + 0.5260 \Rightarrow 0.9044$ bits.

$H(X,Y) = 0.5211 + 0.1129 + 0.5142 + 0.3322 + 0.4644$

$= 2.2770$ bits.

$H(Y/X) = H(Y) - H(Y/X) = 0.9044 - 0.6994$

$= 0.2050$ bits

Cross-Entropy:

$H(P_X, \phi) = -\sum P_X \log_2 Q = 0.32(1.000) +$

$0.38(1.7370) + 0.30(2.3219) = 1.6786$ bits.

- KL divergence:

$D_{KL}(P_X \| Q) = H(P_X, Q) - H(X)$

$= 1.6766 - 1.5776 = 0.0990$ bits

$P(Y=yes / X=High) = 0.20/0.30$

$= 0.6667.$

(Low = 0.0625, medium = 0.2632).

Interpretation & Conclusion:

The mutual information of 0.205 bit is positive, so load carries useful but only = 0.0990 bits.

Low load (0.5 vs 0.32) & High Load (0.2 vs 0.30).

3. i) Given data:

samples $x = \{48, 52, 55, 47, 53, 51\}$

$n = 6.$

Numerical Soln:

ML mean: $\mu = 306/6 = 51.0°C.$

squared deviations:

$(-3)^2 + 1^2 + 4^2 + (-4)^2 + 2^2 + 0^2 = 9 + 1 + 16 + 16 + 4 + 0$

$= 46.$

ML variance $\sigma^2 = 46/6 = 7.667.$

$\sigma = 2.769°C.$

Unbiased variance ; $s^2 = 46/5 = 9.2$

$s = 3.033°C$.

$P(X > 55)$: $z = (55-51)/2.769 = 1.445$

$P = 1 - \phi(1.445) = 0.0743$

$P(48 < X < 54)$: $z = \pm(3/2.769) = \pm 1.083$;

$P = 2\phi(1.083) - 1 = 0.7214 \approx 0.72$.

Accept 0.72 ± 0.01 using $\phi(1.08) = 0.8599$.

ii) Given data:

$(x, y) = (10, 25), (20, 38), (30, 52), (40, 61), (50, 79)$

ridge (λ) = 500 on centred data.

Numerical Soln:-

means: $\bar{x} = 30$, $\bar{y} = (25+38+52+61+77)/5 = 51$

$S\text{-}xx = \Sigma(x-\bar{x})^2 = 1,000$

$S\text{-}xy = \Sigma(x-\bar{x})(y-\bar{y}) = 1,310.$

$w_1 = 1,310/1000 = 1.31$ ms per %.

$w_0 = 51 - 1.31 \times 30 = 11.7$ ms.

Fitted values: 24.8, 37.9, 51.0, 64.1, 77.2,

residuals: 0.2, 0.1, 1.0, −3.1, 1.8

$SSE = 0.04 + 0.01 + 1.00 + 9.61 + 3.24 = 13.90.$

SST = 676+169+1+100+784 = 1730,

R^2 = 1 − 13.90 / 1730

= 0.9920.

Prediction at x = 60, $\bar{y}$ = 11.7 + 1.31 × 60

= 90.3 ms

Ridge: $w_1 = S_{xy}/(S_{xx}+1)$ = 1310/1500

= 0.8733

w_0 = 57 − 0.8733 × 30 = 24.8;

$\hat{y}(60)$ = 24.8 + 52.4 = 77.2 ms

SSE = 204.58, R^2 = 0.8817.

4. Given;

N-spam = 40, N-Legit = 60;

Counts (spam, legit); offer (28, 6) =

free (24, 3), meeting (4, 33); Laplace

smoothing $P(w=1/c) = (n+1)/(N_c+2)$;

Cost of FP = 5, FN = 1.

TP = 45, FN = 15, FP = 10, TN = 130.

Numerical soln:

prior: P(Spam) = 40/100 = 0.40,

P(legit) = 60/100 = 0.60.

Smoothed likelihood, Spam(42).

offer (29/42) = 0.6905, free 25/42 = 0.5952,

meeting 5/42 = 0.1190.

Smoothed, legitimate(62); offer 7/62 = 0.1129

free 4/62 = 0.0645; meeting 34/62 = 0.5848.

Scores for x = (offer = 1, free = 1, meeting = 0),

spam = 0.40 × 0.6905 × 0.5952 × (1 − 0.1190) = 0.1448.

Legit = 0.60 × 0.1129 × 0.0645 × (1 − 0.5484) = 0.00197

Posterior:

P(Spam/x) = 0.1448/(0.1448 + 0.00197)

= P(legit/x) = 0.0314.

Accuracy = (45 + 130)/200 = 87.5%.

precision = 45/(45 + 10) = 81.82%.

Recall = 45/(45 + 15) = 75.0%.

F1 = 2PR/(R + P) = 78.26%.

Total misclassification cost = 5 × FP + 1 × FN

= 5 × 10 + 15

= 65 units

False (+) rate = 10/140 = 7.14%.

CA:

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$\psi_c(x_c)$ =

5. Given,

$x = (0.5, 1.0)$, $\eta = 0.5$

$\sigma(z) = 1/(1 + e^{-z})$,

$E = 1/2 (t-y)^2$.

$w_{11} = 0.4$ & $w_{21} = -0.2$,

$b_1 = 0.1$; $w_{12} = 0.3$

$w_{22} = 0.5$, $b_2 = -0.1$,

$v_1 = 0.6$, $v_2 = -0.4$, $b_3 = 0.2$.

$\therefore z_1 = 0.4(0.5) + (-0.2)(1.0) + 0.1 = 100$

$z_2 = 0.3(0.5) + 0.5(10) - 0.1 = 0.550.$

$h_1 = \sigma(0.100) = 0.5250$

$h_2 = \sigma(0.550) = 0.6341$

output $z_0 = 0.60(0.5250) - 0.4(0.6341) + 0.2$

$= 0.2613.$

$y = \sigma(0.2613) = 0.5650.$

Error: $E = 1/2 (1 - 0.5650)^2 = 1/2 (0.4350)^2$

$= 0.0946.$

Forward value:

$E = 1/2 (0.4122)^2$

$= 0.0850.$